Weekly Diary – Week 8

Internship Organization: Vikaspedia

Duration: 08/03/2026 – 14/03/2026

Objective of the Week

The main objective for this week was to implement real-time communication within the system using WebSockets and to ensure seamless synchronization of document changes among multiple users. The focus was on enabling instant data exchange so that edits made by one user are immediately reflected to others. This involved strengthening both frontend and backend communication mechanisms and improving the efficiency of data transmission.

Date – 8 March

On this day, WebSocket functionality was successfully integrated into the frontend using an appropriate client-side library. The system was configured in such a way that whenever a user opens the editor interface, a WebSocket connection is automatically initiated. This ensured that users could instantly connect to the server without requiring any manual setup, thereby improving usability and responsiveness.

Date – 9 March

The backend development focused on creating WebSocket handlers capable of processing incoming messages from connected clients. These handlers were designed to efficiently manage multiple active connections at the same time. Additional logic was implemented to ensure that updates received from one client could be handled and redistributed properly, forming the backbone of real-time communication.

Date – 10 March

A structured message format was designed and implemented for transmitting editing operations between users. This format defined how different types of changes, such as text insertion and deletion, would be communicated. Extensive testing was carried out to ensure that messages were correctly sent, received, and interpreted by both the frontend and backend systems.

Date – 11 March

The next step involved implementing a broadcasting mechanism. This allowed updates made by one user to be instantly transmitted to all other connected users. Special attention was given to reducing latency so that changes appear almost immediately across all clients. This feature played a crucial role in achieving true real-time collaboration.

Date – 12 March

Further improvements were made to the synchronization logic to handle rapid and continuous edits. Situations involving overlapping or conflicting updates were carefully analyzed and addressed. Adjustments were made to ensure that the document state remained consistent across all users, even during high-frequency editing scenarios.

Date – 13 March

The system was thoroughly tested by opening multiple browser tabs and using different devices simultaneously. These tests helped verify that real-time updates were consistently reflected across all instances without requiring manual page refresh. The results confirmed that synchronization was functioning reliably under different usage conditions.

Date – 14 March

On the final day, debugging and performance optimization were carried out. Issues such as duplicate message delivery, delayed responses, and minor inconsistencies in synchronization were identified and resolved. The communication flow between client and server was further optimized to ensure smoother performance and improved efficiency.